# HydroGate: Evidence-Bounded Maintenance Decisions for a Multi-Sensor Hydraulic Test Rig

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ECE 532 Final Project Proposal, graduate track • Data: UCI #447 (CC BY 4.0), access confirmed • CPU only
Repository: https://github.com/1094357992/532

**Abstract**

Condition-monitoring classifiers act on every input, so a drifted pressure transducer yields a confident, wrong instruction to keep a degrading machine running. We propose **HydroGate**, a CPU-only system for a public 17-sensor hydraulic rig. It emits typed component-condition claims with provenance and risk tiers and acts only when tier-specific obligations pass: signal integrity, cross-modal physical balance, calibrated confidence, a singleton conformal set, and a causal signature. Otherwise it repairs, requests evidence, hands off, or defers, logging each decision in a course schema-1.0 receipt. We ask whether classifier-independent checkers bound false admissions under sensor faults better than uncertainty-only selective prediction. In a completed smoke run (measured, one seed), injected sensor faults left a conformal baseline admitting up to 53.3% wrong claims, whereas the gate with repair stayed at $\leq$4.1% (clean: 3.7%) while admitting 53–63% of claims.

## 1 Problem & Research Question

UCI dataset #447 (Helwig et al., 2015) holds 2,205 sixty-second cycles of a rig whose cooler, valve, pump, and accumulator are degraded in controlled grades, with 17 channels (pressure, motor power, flow, temperature, vibration, virtual efficiency) at 1–100 Hz.

**Decision boundary.** Per cycle and component, the system returns `act`, licensing CONTINUE (healthy), SCHEDULE (moderate degradation), or STOP (near failure), or another course outcome: `repair` (drop a faulty channel), `request_evidence` (observe the next cycle), `hand_off` (technician), or `defer` (a check is unavailable). The costly error is a *false admission*, such as CONTINUE on a pump with severe leakage.

**Why it matters.** Cooler and valve states are classified almost perfectly (Helwig et al., 2015; Schneider et al., 2017), but sensors drift, stick, and drop out, which degrades classifiers on this rig (Helwig and Schütze, 2015). In our smoke run, a $3\sigma$ offset on one pressure channel raises the ungated false-admission rate (FAR) from 15.4% to 56.1%.

**Research question.** *Does a fail-closed, risk-tiered gate built on classifier-independent checkers (per-channel integrity C1; cross-modal physical balance C2) reduce false admissions under sensor faults and operating shift, relative to confidence- and conformal-only selective prediction, at bounded cost in clean-cycle deferral?*

## 2 Prior Work & Contribution

Helwig et al. (2015) and Schneider et al. (2017) classify this rig's states (pump and accumulator are hardest) and act on every prediction. Helwig and Schütze (2015) compensate offset and drift at the feature level, without abstention or an audit trail. Selective classifiers abstain below a risk-calibrated confidence threshold (Geifman and El-Yaniv, 2017), and split-conformal sets give coverage under exchangeability (Angelopoulos and Bates, 2023); both abstain on *model* uncertainty. A corrupted channel breaks exchangeability and yields *confident* errors (B2 in Table 1).

**Gap and contribution.** None of these ties an action to the validity of its sensor evidence, scaled by risk, in a re-checkable record. We contribute (i) typed claims with provenance, risk tiers, and obligations in the course schema (observation → claim → obligation → checker → warrant → gate → receipt), consuming and extending A1's representation-evidence record and A2's calibrated evidence bundle and act/acquire/defer policy; (ii) two independent checkers with fault localization, plus causal signatures validated on the rig's designed experiment; (iii) a fail-closed gate whose invariant is re-verified from the trace; and (iv) a failure-injection benchmark.

## 3 Technical Approach

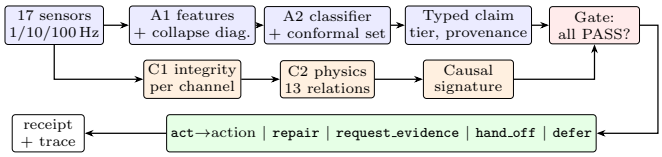



Figure 1: HydroGate. Checkers (orange) read raw signals and share no parameters with the classifier path (blue). `repair` re-runs without the localized channel.

**A1: representation evidence.** All 20 data files are SHA-256 pinned (loading fails closed). Each channel-cycle becomes six segment means plus std, min, max, and slope: 170 frozen features. As in A1, the record holds an artifact ID and provenance, configuration, supported claim, collapse diagnostics (effective rank and per-dimension std of row-normalized features), bounded conditions, and a fail-closed disposition. Training effective rank is 23.6; per evaluation batch, a stuck pressure channel drops it to 5.0 and a dropout leaves 10 dead dimensions.

**A2: calibrated evidence and policy.** Per com-

ponent, logistic regression or gradient-boosted trees give probabilities and split-conformal LAC sets at $\alpha$=0.05 (Angelopoulos and Bates, 2023); the A2 bundle records ECE ($B$=15), NLL, $\hat{q}$, coverage, and available channels. We extend A2's single $\theta_{\text{act}}$=0.85 to tier-specific thresholds and map its *acquire* to `request_evidence`. Repair reuses A2's availability mask: a faulty channel is masked and the claim re-derived by a drop-one-channel model.

**Claims and obligations.** A component-condition claim carries grade, confidence, conformal set, and provenance (data hash, feature spec, model digest). Its course risk tier is the cost of a wrong admission: **medium** (moderate → SCHEDULE), **high** (healthy → CONTINUE), **critical** (near failure → STOP). The obligations are: O1, C1 passes on every channel used; O2, all C2 relations hold; O3, confidence $\geq \tau_T$ (0.80/0.90/0.95); O4, the causal signature matches the grade; O5, the conformal set is a singleton. Medium requires {O1, O3}, high {O1, O2, O3, O5}, and critical all five.

**Checkers.** *C1* tests each channel alone (non-finite, range, flatline, jumps) against training thresholds. *C2* fits 13 linear balance relations on cycle means, e.g. motor power on pressure, flow, their product, and temperature ($R^2$=0.993), plus redundant pairs and a PS1–PS2 coherence band. A relation fails when its residual exceeds 1.5× its training 99.5th percentile, and voting over failing and passing relations localizes the fault. *Causal signatures* are signed degradation effects, such as *pump leakage*↑⇒ *flow FS1*↓, kept only with sign agreement in ≥90% of strata with other components fixed (the rig's designed experiment). Measured support is 1.00 (cooler→CE), 1.00 (pump→FS1), and 0.95 (valve→early FS1); validated claims become per-grade intervals. No accumulator signature is identified, so critical accumulator claims fail O4 and never act.

**Gate, policy, receipt.** A claim acts only if every obligation of its tier has an executed `PASS`. As in the course `evaluate_gate`, an unavailable or errored check yields `defer`. Otherwise, one localized channel yields `repair` (re-checked once), failures of only O3/O5 on medium or high claims yield `request_evidence`, and anything else yields `hand_off`. The **invariant** is *no action without an executed PASS on every obligation of its tier*. It is checked at runtime, in exhaustive tests, and from the trace. Each run emits a schema-1.0 `project` receipt that passes the course validator (`--strict`), plus a hash-pinned audit and decision trace.

# 4 Evaluation

**Hypotheses (expected outcomes).** *H1:* under single-channel faults, the full gate G has lower FAR than conformal selective prediction B2, with clean unnecessary rejection ≤5 points above B2's. *H2:* repair (G+R) restores coverage to within 10 points of clean at no more than clean FAR. *H3:* removing C1 or C2 raises FAR on the faults it covers.

**Baselines and metrics.** B0 always acts; B1 uses tiered confidence thresholds; B2 adds a conformal singleton; ablations remove C1, C2, or the signature check. Task metrics (accuracy, macro-F1, ECE, NLL) are reported apart from policy metrics (coverage, FAR, selective risk, unnecessary rejection, critical misses) and checker metrics (channel coverage C1 17/17, C2 14/17; detection; clean false alarms; repair success), plus runtime and peak RSS.

**Partitions.** Contiguous 30-cycle blocks go 60/20/20 to train, calibration, and test, so near-identical neighbors never straddle partitions; only stable cycles are used. Unstable test-block cycles form a shift partition, and the 73% valve grade is held out as novel severity; 5 seeds × 2 models.

**Failure tests.** Dropout, stuck-at, 3$\sigma$ offset, +5% gain, 10% time shift, and spikes are injected on seven channels. A checker-aware adversary applies the smallest bias that flips ≥25% of ungated predictions while passing C1, which isolates C2. Seven fabricated claims (e.g., remaining useful life, missing provenance) must be rejected; the receipt lists unwarranted claims.

**Support or contradiction.** H1 holds if G's FAR is below B2's for every fault mode and seed within the clean bound. It fails if the checkers fire mainly where B2 already rejects, if clean false alarms exceed 5%, or if checker-passing biases push FAR toward B0. Three partial contradictions are measured. (a) On unstable cycles C2 fires on 17.1% and G's FAR stays 20.7% (Table 1): physics-preserving shift escapes the gate. (b) Under FS1 dropout, repair misses H2 by 11 coverage points. (c) Medium claims skip O2, so a C1-passing adversarial bias that C2 flags on 100% of cycles still yields 20.6% FAR (B2: 34.1%), and the held-out valve severity 26.9%.

# 5 Feasibility & Milestones

**Completed (measured 6 Nov 2026).** Data access is confirmed; smoke runs end to end on 735 cycles in 6.8 s (655 MiB). All 8 tests pass; the smoke receipt (28 claims, 65 checks, 28 gates) passes the course validator in strict mode, with 0 invariant violations and 0/7 fabricated claims admitted. The full `cpu.yaml` run (5 seeds × 2 models) took 24 min on 2 cores at about 1.0 GiB per process (budget: 2 h), and validates with 0 violations over 384,820 decisions. Averaged over seeds and channels (logistic regression), G holds FAR at ≤4.8% under dropout, stuck, spike, offset, and gain faults, versus 7.4–34.8% for B2, with clean false alarms of 0.5% (C1) and 0.4% (C2).

| Condition | B0 FAR | B2 FAR | G FAR / cov | G+R FAR / cov | rep. |
|---|---|---|---|---|---|
| Clean | 15.4 | 3.7 | 3.7 / 64.1 | 3.7 / 64.1 | – |
| Offset 3$\sigma$ PS1 | 56.1 | 34.3 | 0.0 / 0.0 | 3.7 / 61.5 | 57.8 |
| Dropout FS1 | 22.4 | 6.1 | 0.0 / 0.0 | 4.1 / 52.8 | 48.7 |
| Unstable (shift) | 37.9 | 29.3 | 20.7 / 50.0 | 30.7 / 60.0 | 0.0 |

Table 1: Smoke results (measured; one seed; %). cov: admitted; rep.: repair success.

**Planned.** C2 veto for all tiers, ablations (Nov 13); accumulator signature, repair guard (Nov 20); progress gate, lockfile (Dec 1); poster (Dec 11); report and reproducibility appendix (Dec 15).

**Risks and fallbacks.** (1) Weak accumulator model (41.7% vs. 97–100%): allow STOP for three components only. (2) Repair hurts under shift (FAR 20.7→30.7%): repair only inside the unrepaired conformal set. (3) PS3, PS4, VS1 lack C2 coverage. **Owner:** Yilong Peng; no scale-up.

# Annotated References

A. N. Angelopoulos and S. Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101. *Annotation:* Tutorial source for the split-conformal LAC procedure we use in A2 (score $1 - \hat{p}_y$, finite-sample quantile $\hat{q}$). It also states the exchangeability assumption that sensor faults violate, which motivates evidence checkers that act independently of the conformal set.

Y. Geifman and R. El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30*, pages 4878–4887, 2017. *Annotation:* Defines selective classification by risk and coverage, with a confidence-threshold rejection function. Our B1 baseline and our coverage, selective-risk, and false-admission metrics follow this risk–coverage view.

N. Helwig and A. Schütze. Detecting and compensating sensor faults in a hydraulic condition monitoring system. In *Proc. SENSOR 2015, 17th Int. Conf. on Sensors and Measurement Technology*, Nuremberg, 2015. doi: 10.5162/sensor2015/D8.1. *Annotation:* The closest prior work. It studies offset and drift faults on this rig and compensates them at the feature level. We instead detect and localize faults with classifier-independent checkers, decline or repair at the decision level, and record the outcome, and we use its fault types to design our injection suite.

N. Helwig, E. Pignanelli, and A. Schütze. Condition monitoring of a complex hydraulic system using multivariate statistics. In *Proc. IEEE I2MTC*, Pisa, Italy, 2015. doi: 10.1109/I2MTC.2015.7151267. *Annotation:* Describes the test rig, its sensors, and the controlled degradation grades of UCI dataset #447, our data source. It reports that cooler and valve states are separable while pump and accumulator states are harder, which matches our smoke results and our accumulator fallback.

T. Schneider, N. Helwig, and A. Schütze. Automatic feature extraction and selection for classification of cyclical time series data. *tm – Technisches Messen*, 84(3):198–206, 2017. doi: 10.1515/teme-2016-0072. *Annotation:* Automated extraction of cycle-shape, frequency, and time-frequency features, with selection, evaluated on this dataset among others. It is the accuracy-oriented reference point for our frozen A1 features. Its richer features are a planned A1 extension for the accumulator.